Lab 4: Trajectory Optimization for UAVs

Minimum-Snap Planning and Drone Racing in ROS/TESSE

Group 14

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Individual work: Longchuan Zheng

Abstract

The VNAV Lab 4 implementation generates a minimum-snap trajectory through ordered gate poses and tracks it with a motor-driven quadrotor in ROS Noetic and TESSE. The planner combines continuous segment-time allocation, bounded local retiming and a smooth execution clock. The controller adds delayed attitude feedforward. The fastest complete lap was 7.44 s, with all twelve gates passed in order. Successful fast trials were approximately 7.44–7.58 s across the tested settings and runs. A representative saved run, `path110.bag`, is used for tracking analysis; its sampled reference peak speed was 13.57 m/s, with peak acceleration of 25.12 m/s^2. The recorded flight is used to analyze path tracking, altitude, velocity and motor commands. The report combines Longchuan Zheng's solutions to Individual Deliverables 1 and 2 with the team's implementation and experimental results for Deliverable 3.

1 Objectives

Deliverable 3 requires gate publication, trajectory generation, desired-state sampling, controller integration and a recorded race [1]. The individual exercises develop the polynomial optimization and constraint counting used in trajectory planning. The experimental sections cover the team's motor-driven implementation and results. The best complete lap was 7.44 s. The plots and tracking metrics use the representative `path110.bag` run associated with a 7.58 s timer reading; this recording is distinct from the fastest trial.

1.1 Team Contributions

The work was divided into four technical areas. Each member also prepared the corresponding report material.

Member	Lab 4 contribution
Yuanxi Han	Trajectory generation and time allocation: continuous trajectories, velocity and acceleration constraints, trajectory-time compression, and analysis of reference feasibility.
Longchuan Zheng	Trajectory tracking and attitude control: position and attitude error analysis, controller-parameter and attitude-feedforward adjustment, and evaluation of the high-speed tracking response.
Zhiran Wang	Waypoint and course geometry: verification of gate positions, coordinate conversion and ordering, analysis of sharp turns and the final gate, and local path-adjustment proposals.
Haokuan Li	Experimental evaluation and reproducibility: parameter comparison experiments, rosbag recording and analysis, assessment of time, gate passage and contacts, and preparation of the reproducible environment.

2 Individual Deliverable 1: Single-Segment Optimization

Individual solutions by Longchuan Zheng.

Polynomial coefficients are listed in increasing powers, with segment durations held fixed.

Let $P(0) = 0$, $P(1) = 1$ and $J = \int_0^1 (P'(t))^2 dt$. For $P(t) = \sum_{j=1}^n p_j t^j$,

$$J = \boldsymbol{p}^\top Q \boldsymbol{p}, \qquad Q_{ij} = \frac{ij}{i+j-1}, \qquad A = \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix}, \quad b = 1. \tag{1}$$

If the QP solver uses $\frac{1}{2}\boldsymbol{p}^\top H \boldsymbol{p}$ as its objective, set $H = 2Q$.

2.1 Question 1: Degree One

$P(t) = p_1 t$ gives $p_1 = 1$, $P^\star = t$ and $J^\star = 1$.

2.2 Question 2: Degree Two

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 4/3 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad b = 1. \tag{2}$$

Substituting $p_1 = 1 - p_2$ gives $J = 1 + p_2^2/3$, minimized at $(p_1, p_2) = (1, 0)$ with $J^\star = 1$. Thus $P^\star(t) = t$; allowing a quadratic term does not reduce the cost.

2.3 Question 3: Degree Three and Euler–Lagrange Consistency

$$Q = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4/3 & 3/2 \\ 1 & 3/2 & 9/5 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \quad b = 1. \tag{3}$$

For nonzero $\boldsymbol{p}$, the polynomial derivative is nonzero, so the integral $\boldsymbol{p}^\top Q \boldsymbol{p}$ is strictly positive. Hence the matrix is positive definite and the feasible QP has a unique minimizer. Cauchy–Schwarz gives $J \geq (\int_0^1 P' dt)^2 = 1$, with equality at $P' = 1$. Thus $\boldsymbol{p}^\star = (1, 0, 0)^\top$. This agrees with the Euler–Lagrange equation $P'' = 0$. Allowing higher-degree terms leaves the minimum unchanged.

2.4 Question 4: Additional Terminal Velocity $P'(1) = -2$

For degree two,

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \tag{4}$$

The unique feasible polynomial is $P = 4t - 3t^2$, with $J = 4$. For degree three, retain the preceding cubic Q and use

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \tag{5}$$

Set $p_3 = c$, $p_2 = -3 - 2c$ and $p_1 = 4 + c$. Then

$$J(c) = 4 + c + \frac{2c^2}{15}, \qquad c^\star = -\frac{15}{4}, \tag{6}$$

$$P^\star(t) = \frac{1}{4}t + \frac{9}{2}t^2 - \frac{15}{4}t^3, \qquad J^\star = \frac{17}{8}. \tag{7}$$

The result is the minimum over the specified cubic family. Since the terminal velocity constraint excludes $P = t$, allowing a higher polynomial degree now changes the minimum cost.

3 Individual Deliverable 2: Multi-Segment Constraints

For fixed segment durations, consider the objective

$$J_r = \sum_{i=1}^{k} \int_0^{T_i} (P_i^{(r)}(t))^2 dt. \tag{8}$$

Assume specified waypoint positions, fixed start/end derivatives of orders $1, \ldots, r-1$, and free shared derivatives at intermediate waypoints. The Euler–Lagrange equation $P_i^{(2r)} = 0$ gives a segment degree of at most $2r-1$, or $2rk$ scalar coefficients per coordinate.

3.1 Question 1: Minimum Snap, $r = 4$

The degree-seven segments have $8k$ coefficients in total. Specifying position at both ends of each segment gives $2k$ equations, with two equations for each interior waypoint. Fixing endpoint velocity, acceleration and jerk adds six equations. Continuity of these three derivatives adds $3(k-1)$ essential equations, leaving $3(k-1)$ unknowns that can be expressed as free shared knot derivatives.

At a fixed intermediate position, $\delta P = 0$, while $\delta P'$, $\delta P''$ and $\delta P'''$ remain free. Stationarity requires natural matching of the derivatives of orders 4, 5 and 6. This adds $3(k-1)$ equations, giving

$$2k + 6 + 3(k-1) + 3(k-1) = 8k. \tag{9}$$

For $k = 3$, this gives $6 + 6 + 6 + 6 = 24$ equations. The natural matching conditions arise when the free derivatives are optimized; they are not additional boundary values chosen in advance.

3.2 Question 2: General Derivative Order r

The general count is

$$\underbrace{2k}_{\text{positions}} + \underbrace{2(r-1)}_{\text{fixed endpoint derivatives}} + \underbrace{(r-1)(k-1)}_{\text{essential continuity}} + \underbrace{(r-1)(k-1)}_{\text{natural matching}} = 2rk. \tag{10}$$

The natural matching conditions apply to derivatives of orders $r, \ldots, 2r-2$. There are $(r-1)(k-1)$ free shared intermediate derivatives. Changing the boundary conditions or optimizing the durations changes this count.

3.3 Constraint Classification and Counting Convention

The counts below are scalar and apply independently to each spatial coordinate. At an interior waypoint, a specified position constrains both adjacent segment endpoints. A free shared derivative contributes one continuity equation and, at the optimum, one natural matching equation. The matching equation comes from stationarity of the cost, while the shared derivative remains free to be optimized. With the stated endpoint conditions, the count is

Condition	Minimum snap	Order r
Waypoint positions	$2k$	$2k$
Fixed endpoint derivatives	6	$2(r-1)$
Continuity of free knot derivatives	$3(k-1)$	$(r-1)(k-1)$
Natural matching at free knots	$3(k-1)$	$(r-1)(k-1)$
Total equations	$8k$	$2rk$

There are $3(k-1)$ free shared knot derivative values for minimum snap, and $(r-1)(k-1)$ in the general case. In a QP formulation they remain optimization variables; the cost determines their optimal values. They should not be counted as additional prescribed derivative constraints.

Integrating by parts shows where these boundary conditions come from:

$$\frac{\delta J_r}{2} = \sum_i \left[\sum_{j=0}^{r-1} (-1)^j P_i^{(r+j)} \delta P_i^{(r-1-j)} \right]_0^{T_i} + (-1)^r \sum_i \int_0^{T_i} P_i^{(2r)} \delta P_i \, dt. \tag{11}$$

The integral yields $P_i^{(2r)} = 0$. At a fixed interior position, $\delta P_i = 0$, so no matching of derivative order $2r-1$ is imposed. Independent shared variations of orders $1, \ldots, r-1$ instead yield matching of orders $r, \ldots, 2r-2$, as counted above.

The degree derived above is an upper bound; particular problems may have optimal segments of lower degree. For $r = 1$ there are no shared derivative variables or natural matching conditions; fixed waypoint positions determine each linear segment.

4 Experimental Setup

4.1 Software and Data Flow

The exported experiment environment was Ubuntu 20.04.6 LTS (x86-64) in VMware, with ROS Noetic, Python 3.8.10, catkin tools and the original motor-driven Lab 4 TESSE simulator. The reproduction package records the simulator checksum and installed software versions. Positive world-frame z points upwards. The TESSE bridge forwards the controller's four rotor-speed commands to the simulated vehicle.

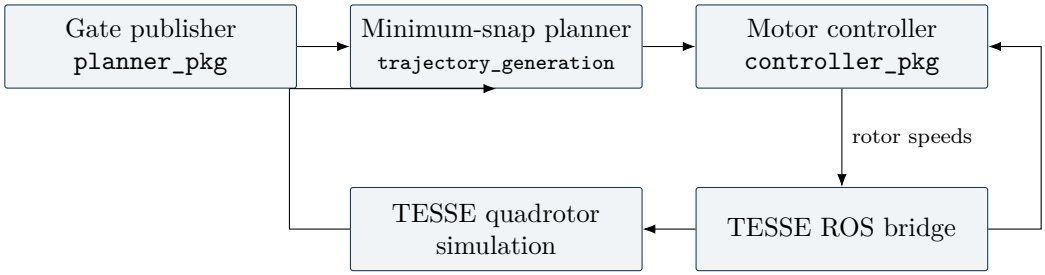


Figure 1: Data flow during the race. The bridge supplies measured state to the planner and controller. The planner publishes desired position, velocity and acceleration, together with a companion attitude-feedforward message.

4.2 Configuration and Experimental Procedure

The saved path110 configuration used $(v_{\max}, a_{\max}, a_{\text{lat}}) = (21, 16, 8)$, `time_scale=1.27`, a first-gate vertical-speed cap of 3 m/s, a retiming target ratio of 0.93 with at most twelve passes, and `path_speedup=1.10` with a 1.0 s ramp. The inspected launch file uses $(k_x, k_v, k_R, k_\Omega, k_{x,z}, k_{v,z}) = (1, 2, 4, 1, 4, 4)$. Attitude feedforward uses gain 0.25, start time 2.0 s, ramp duration 0.5 s and torque cap 2. Identified inertia and learning are disabled. These settings are explained in the planning and control sections below.

Before a run, the vehicle is reset and the simulator, bridge and external timer are started. The follower and recorder are made ready before gate publication triggers trajectory generation. Installation commands, terminal entry points, package checksums and repeat-run instructions are provided in the reproduction package's README.

5 Gate Publication and Minimum-Snap Planning

5.1 Part 0 and the Planner Entry Points

In `simple_traj_planner.cpp`, the warm-up reference is the first gate's position and orientation. The node publishes this desired state with zero-initialized velocity and acceleration and broadcasts its transform, as requested in Part 0. For the race, Part 1.1 reads the measured initial state, Part 1.2 solves the trajectory, and Part 1.3 samples it on a requested 0.01 s timer.

5.2 Gate Order and Coordinates

The publisher reads `vnav-2020-lab4_static_tfs.csv` from the simulator's `StreamingAssets` directory and selects the `red_square_drone_door` objects. It converts their poses from Unity to ROS coordinates, sorts them by numeric suffix, and publishes the ordered list as a `PoseArray` on `/desired_traj_vertices`. The object without a suffix has map index zero. It is labelled Gate 1 here, giving twelve gates in total.

5.3 Polynomial Path and Boundary Conditions

The generator uses the measured initial position as the first waypoint, followed by the gate centers. Each segment has ten coefficients per coordinate,

$$\boldsymbol{p}_i(t) = \sum_{j=0}^{9} \boldsymbol{c}_{ij} t^j, \quad 0 \le t \le T_i, \qquad \min \sum_i \int_0^{T_i} \|\boldsymbol{p}_i^{(4)}(t)\|^2 dt. \tag{12}$$

The solver is `PolynomialOptimization<10>` with the `SNAP` objective. Start and end vertices are set with `makeStartOrEnd(..., SNAP)`. Intermediate vertices fix position, and the optimization library shares derivatives at segment boundaries. After the first solve, the first-gate reference vertical velocity is capped above at 3 m/s and the problem is solved again. Reference yaw stays at the measured initial yaw.

The reference follows a smooth polynomial through the gate centers. Straight-line distances between waypoints are used to estimate segment durations. The source's straight-line log message therefore describes the timing estimate imprecisely when applied to the sampled path.

5.4 Continuous Time Allocation

For segment length d_i, the legacy rest-to-rest duration estimate is

$$\hat{T}_i = \begin{cases} 2\sqrt{d_i/a_{\max}}, & d_i < v_{\max}^2/a_{\max}, \\ d_i/v_{\max} + v_{\max}/a_{\max}, & \text{otherwise}, \end{cases} \qquad T_i = \max(0.3,\ s_t \hat{T}_i). \tag{13}$$

The recorded run uses `timing_mode=continuous`, which allows nonzero speed estimates at intermediate gates. For a heading change θ_i,

$$r_i = \frac{\min(d_{i-1}, d_i)}{2\sin(\theta_i/2)}, \qquad u_i = \min\{v_{\max}, \sqrt{a_{\text{lat}} r_i}\}. \tag{14}$$

The speed cap is retained where the heading is straight, while the initial and final speed estimates are zero. A forward acceleration pass and a backward braking pass enforce

$$u_{i+1} \le \sqrt{u_i^2 + 2a_{\max} d_i}, \qquad u_i \le \sqrt{u_{i+1}^2 + 2a_{\max} d_i}. \tag{15}$$

For segment entry and exit estimates (u, w), the estimated peak is

$$v_p = \min\{v_{\max}, \sqrt{a_{\max} d_i + (u^2 + w^2)/2}\}. \tag{16}$$

The acceleration, cruise and deceleration times are calculated from this peak, then scaled by $s_t = 1.27$ with a 0.3 s lower bound. The corner-radius rule estimates timing. The resulting polynomial derivatives must still be evaluated separately.

5.5 Bounded Local Retiming

Retiming holds the first two segment durations and their boundary velocity, acceleration, jerk and snap fixed. Each trial shortens one later segment by 2% and solves the polynomial again, keeping its duration above $\max(0.3,\ 0.8T_{i,\text{original}})$. The search ends when it reaches the 0.93 total-duration target, completes a pass without accepting a candidate, or reaches twelve passes.

For an accepted candidate, the peak velocity, acceleration, jerk and snap must stay within 1.10, 1.10, 1.15 and 1.20 times their baseline values. After the protected prefix, sampled path deviation is limited to 0.25 m and sampled nominal tilt may increase by at most 3°. These checks apply to the reference trajectory. Actual tracking clearance and actuator feasibility remain unverified by this screening.

5.6 Smooth Execution Clock

After the first two segments, a one-second ramp raises phase rate from 1 to 1.10. Inside that ramp, with $u = (t - t_0)/T_r$,

$$S(u) = 35u^4 - 84u^5 + 70u^6 - 20u^7, \qquad \dot{h}(t) = 1 + 0.10S(u), \quad T_r = 1\,\text{s}. \tag{17}$$

The first three derivatives of S vanish at both ramp boundaries. Applying the chain rule to $\boldsymbol{p}_d(t) = \boldsymbol{p}(h(t))$ gives

$$\boldsymbol{v}_d = \boldsymbol{p}'\dot{h}, \tag{18}$$

$$\boldsymbol{a}_d = \boldsymbol{p}''\dot{h}^2 + \boldsymbol{p}'\ddot{h}, \tag{19}$$

$$\boldsymbol{j}_d = \boldsymbol{p}'''\dot{h}^3 + 3\boldsymbol{p}''\dot{h}\ddot{h} + \boldsymbol{p}'h^{(3)}. \tag{20}$$

Snap is calculated for attitude feedforward as well. A higher phase rate raises both the speed and acceleration required to follow the path.

6 Motor-Driven Trajectory Following

The controller compares measured and desired position and velocity, $\boldsymbol{e}_x = \boldsymbol{x} - \boldsymbol{x}_d$, $\boldsymbol{e}_v = \boldsymbol{v} - \boldsymbol{v}_d$, and commands

$$\boldsymbol{F}_d = m(\boldsymbol{a}_d - K_x\boldsymbol{e}_x - K_v\boldsymbol{e}_v + g\boldsymbol{e}_3), \quad K_x = \text{diag}(1,1,4), \quad K_v = \text{diag}(2,2,4). \tag{21}$$

The desired body z axis is $\boldsymbol{F}_d/\|\boldsymbol{F}_d\|$. Desired yaw determines the remaining axes of R_d. The controller interprets published angular twist in world coordinates and transforms it into body coordinates. The base torque is

$$\boldsymbol{e}_R = \tfrac{1}{2}(R_d^\top R - R^\top R_d)^\vee, \qquad \boldsymbol{\tau}_{\text{base}} = -4\boldsymbol{e}_R - \boldsymbol{\Omega}. \tag{22}$$

Analytic desired-attitude rate and acceleration feedforward begins after reference time 2.0 s. It uses gain 0.25, a 0.5 s blend and a vector-norm cap of 2 in the controller's torque units. The companion

message must be fresh and aligned with the desired-state phase. The controller model uses its default identity inertia matrix for this configuration.

The thrust command is $f_d = \boldsymbol{F}_d^\top R\boldsymbol{e}_3$. Inverting the X-configuration rotor-allocation matrix, with $c_f = 10^{-3}$, $c_d = 10^{-5}$ and arm parameter $d = 0.3\,\mathrm{m}$, gives the values to which the controller applies a signed square root. The resulting four signed rotor-speed commands are forwarded to TESSE. The controller requests a 100 Hz timer, although simulator and bridge scheduling can be slower.

7 Recorded Race Result and Tracking Analysis

7.1 Race Completion

The quadrotor passed all twelve gates in order. The fastest complete lap was 7.44 s on the external race timer. Successful fast trials fell approximately in the 7.44–7.58 s range across the tested settings and runs; 7.58 s is a representative result rather than the best time. The variation reflects both tuning changes and run-to-run differences, so this range is not a statistical confidence interval for a fixed configuration.

Continuous timing and the smooth phase-rate ramp maintain motion between gates rather than requiring a stop at each waypoint. The analysis below uses the saved `path110.bag` recording from the 7.58 s trial. Its plots and metrics describe that recorded flight and are not measurements of the separate 7.44 s trial.

Ground scraping after the startup roll and wall contact near a sharp turn were tolerated during the aggressive trials; contact impulses offer a possible way to assist upward recovery or turning, although a lap-time benefit from this mechanism was not established by a controlled comparison.

Quantity	Recorded value
Fastest complete lap	7.44 s
Approximate successful fast-trial range	7.44–7.58 s
Representative recorded lap (path110)	7.58 s
Gates passed	12/12
Sampled reference peak speed	13.57 m/s
Sampled reference peak acceleration	25.12 m/s^2
Measured-state evaluation window	0.00–9.54 s, 478 samples
Position RMS / maximum error in this window	2.487 / 4.902 m
Measured peak speed in this window	15.93 m/s
Gain set $(k_x, k_v, k_R, k_\Omega, k_{x,z}, k_{v,z})$	$(1, 2, 4, 1, 4, 4)$

7.2 Path, Altitude and Speed

The numbered markers locate the gate centers and show their order. The traces show the startup and subsequent tracking response over the recorded trajectory evaluation window.

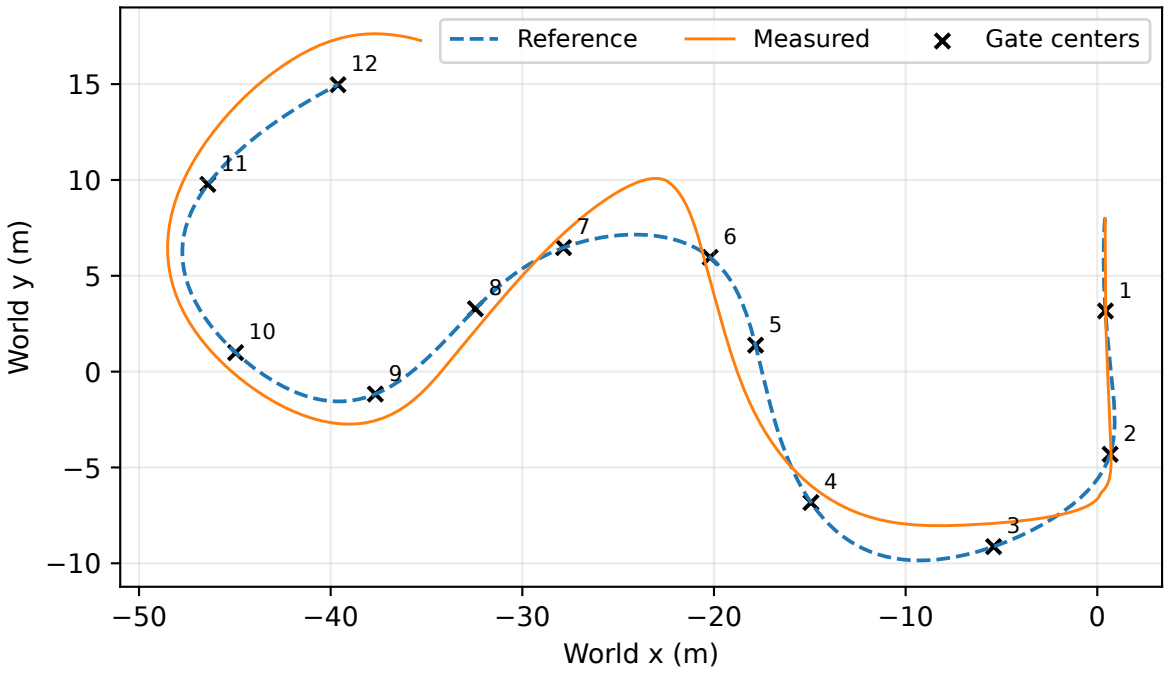


Figure 2: Reference and measured XY paths from `path110.bag`; numbered crosses indicate gate centers.

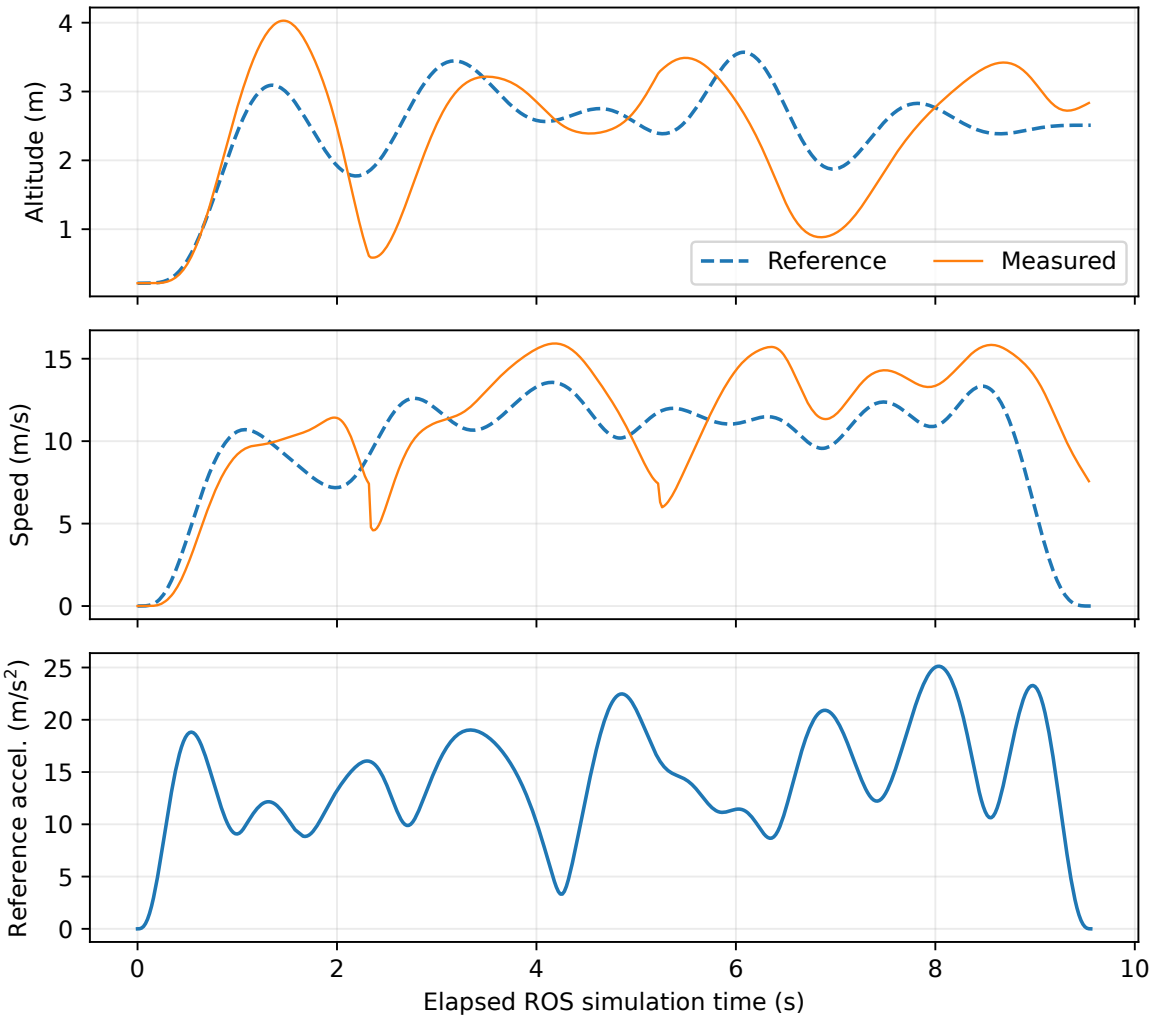


Figure 3: Reference and measured altitude and speed, followed by reference acceleration magnitude. Time is measured in ROS simulation seconds from reference start.

7.3 Tracking Error and Motor Commands

Desired position is linearly interpolated at measured-state timestamps. With $e_p(t) = \|\boldsymbol{x}(t) - \boldsymbol{x}_d(t)\|$, the plotted RMS is

$$e_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} e_p(t_n)^2}. \tag{23}$$

The exact evaluation interval and sample count are saved in `metrics.json`. The error includes the startup transient and subsequent tracking lag. The rotor traces show signed commands. Achieved motor speeds were not measured in this record.

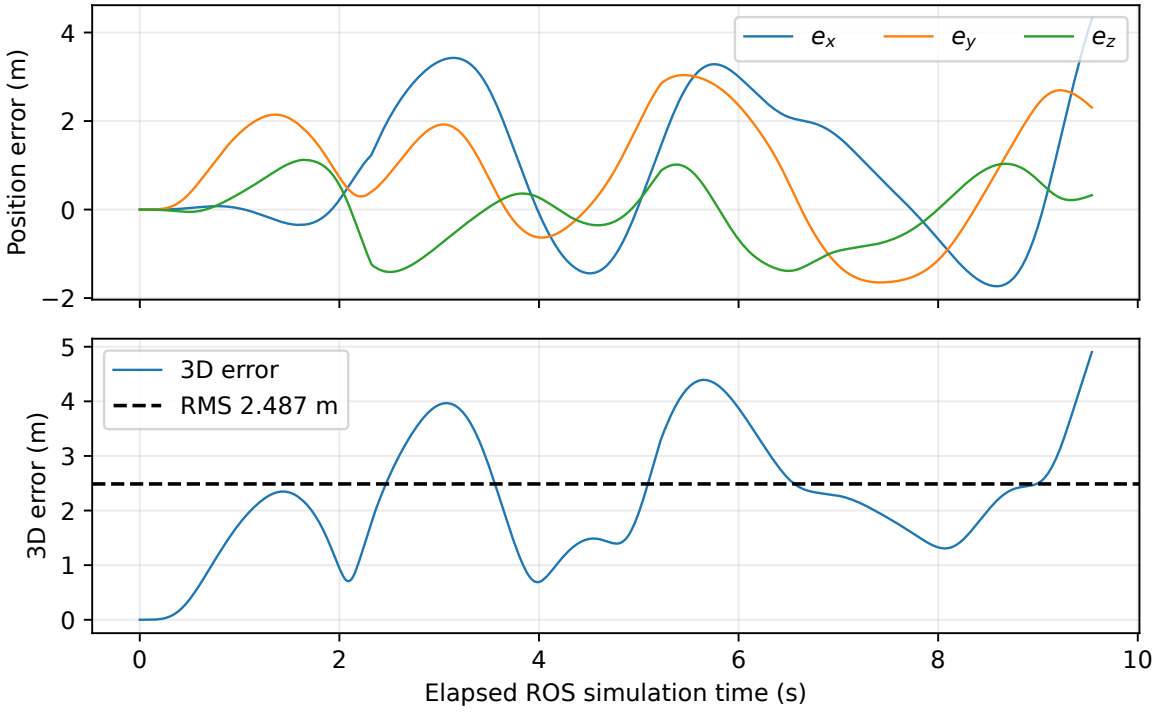


Figure 4: Component-wise and Euclidean position errors with interpolated desired positions at measured-state times.

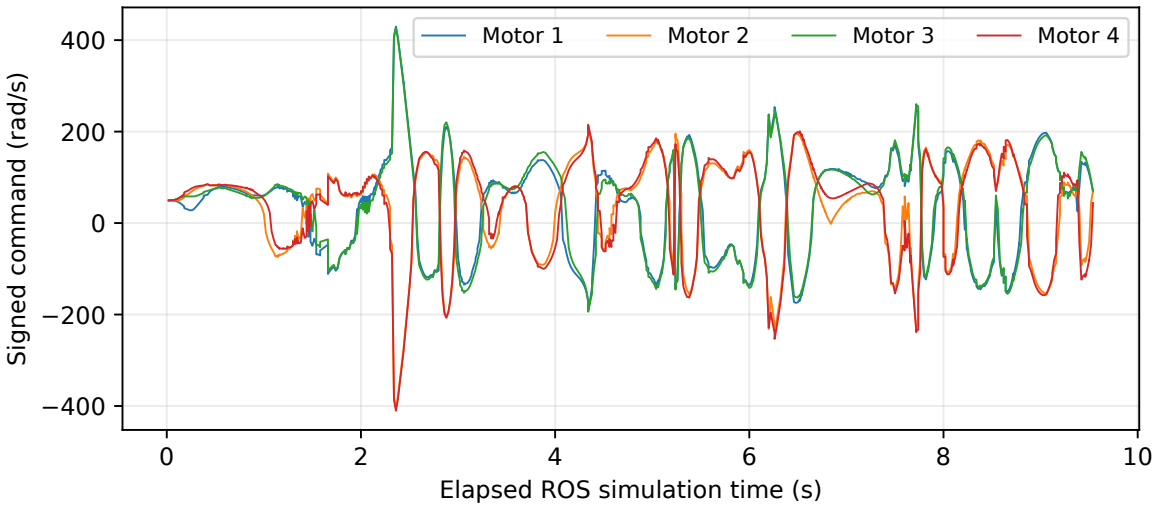


Figure 5: Four signed rotor-speed commands over the planned trajectory interval.

8 Personal Contribution to the Team Experiment

My role in Group 14 was trajectory tracking and attitude control.

My work covered `controller_pkg/src/controller_node.cpp` and its desired-state and attitude-feedforward interfaces. I also analyzed tracking during high-speed flight.

8.1 Position Tracking and Desired Attitude

My controller work included the feedback calculation that converts a reference trajectory into thrust and attitude commands. With position and velocity errors defined as $\boldsymbol{e}_x = \boldsymbol{x} - \boldsymbol{x}_d$ and $\boldsymbol{e}_v = \boldsymbol{v} - \boldsymbol{v}_d$, the requested force is

$$\boldsymbol{F}_d = m(\boldsymbol{a}_d - K_x\boldsymbol{e}_x - K_v\boldsymbol{e}_v + g\boldsymbol{e}_3). \tag{24}$$

I examined how the reference acceleration and feedback correction influence the desired attitude, particularly when the vehicle lags near a turn. In the saved path110 configuration, $K_x = \mathrm{diag}(1, 1, 4)$ and $K_v = \mathrm{diag}(2, 2, 4)$, allowing stronger vertical feedback than horizontal feedback.

8.2 Attitude Feedback and Feedforward Adjustment

I analyzed the attitude and angular-velocity feedback and helped adjust the controller gains and attitude feedforward. The saved configuration uses $k_R = 4$ and $k_\Omega = 1$. The controller transforms measured world-frame angular velocity into body coordinates before applying feedback, and allocates thrust and torque to four signed rotor-speed commands.

To follow rapid attitude changes, the controller adds feedforward based on the desired attitude rate and acceleration. I helped integrate and tune this correction, blending it in gradually to avoid a sudden torque change. The retained gain is 0.25, with a start time of 2.0 s, a 0.5 s ramp and a torque-vector norm cap of 2 in the controller's units. These values set when the correction begins and how strongly it acts.

8.3 Tracking Evaluation and Interpretation

I compared the reference and measured position, altitude and velocity alongside the rotor commands. I used position error to assess recovery after startup and the tracking lag near turns. For the representative `path110.bag` record, I interpolated desired position at the measured-state timestamps. Over the 0.00–9.54 s evaluation window, the position RMS error is 2.487 m and the maximum error is 4.902 m. These metrics include startup and tracking lag.

8.4 Outcome and Scope of My Work

I wrote the sections on the control method and tracking response for the team report. The team passed all twelve gates, achieving a fastest lap of 7.44 s. Successful fast trials were approximately in the 7.44–7.58 s range; the saved path110 recording is from the representative 7.58 s trial. These lap times reflect changes to planning and timing as well as control. They do not measure the effect of the controller adjustments alone.

9 Conclusions

Continuous segment timing, bounded local retiming and a smooth execution clock enabled the motor-driven quadrotor to complete all twelve gates, with a best time of 7.44 s and successful fast trials approximately within 7.44–7.58 s. The recorded data show how the faster reference affects the actual path, velocity and tracking error. Further gains could come from coordinating turn geometry, time allocation and attitude response.

The accompanying reproduction package contains the source snapshot, original simulator and launch scripts, with one README describing installation and execution. The historical recording is provided separately through the download page linked in that README. The exported source snapshot corresponds to the path110 configuration; an exact source revision for the fastest 7.44 s trial has not been identified.

References

[1] MIT VNAV, *Lab 4 Exercises*, Handouts 2023 (ROS 1), supplied course handout.

[2] ETH Zurich, *mav_trajectory_generation*, source distributed with the Lab 4 dependencies.

[3] Group 14, `path110.bag`, saved ROS race record, November 5, 2026; recorded external-timer lap time 7.58 s.